

Name _____ Date _____ Period _____

Parent Functions III: $1/x$ and Asymptotes

Algebra II | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.

Vocabulary

Asymptotic behavior — how a graph behaves near a value it gets closer and closer to but never _____.

Vertical asymptote — a vertical line the graph _____. It sits where the function is undefined.

Horizontal asymptote — a horizontal line the y-values get _____, as x gets very large or very small.

Part 1 — Build the Table for $f(x) = 1/x$

Fill in the outputs. One input has no output at all.

x	-2	-1	-1/2	0	1/2	1	2
$f(x) = 1/x$							

Why is $x = 0$ undefined? _____

Part 2 — What Happens Near Zero and Far Out

Follow the pattern in each list, then describe it.

x	1	0.5	0.1	0.01
$1/x$	1	2	10	100

As x gets close to 0, the outputs grow _____. So there is a vertical asymptote at _____.

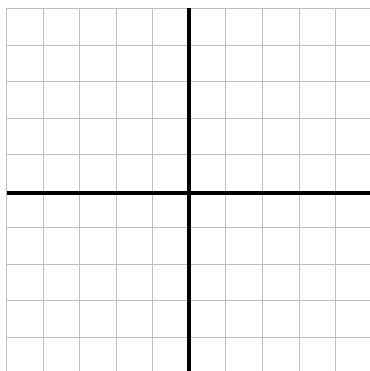
x	1	10	100	1000
$1/x$	1	0.1	0.01	0.001

As x gets very large, the outputs approach _____. So there is a horizontal asymptote at _____.

Part 3 — Graph It

Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y-axis.

$f(x) = 1/x$



Record what you see.

Vertical asymptote: _____

Horizontal asymptote: _____

How many branches? _____

Which quadrants? _____

Any intercepts? _____

Draw both asymptotes as dashed lines.

Part 4 — Key Attributes of $f(x) = 1/x$

Domain: _____ set notation _____

Range: _____ interval _____

Symmetry: reflectional across _____, and rotational _____ about the origin.

Maximum or minimum? _____ Intercepts? _____

This is the first parent function with _____ and the first whose domain has a _____ in it.

Part 5 — True or False

1. A graph may cross a vertical asymptote. _____
2. A graph may cross a horizontal asymptote. _____
3. $f(x) = 1/x$ has a y-intercept. _____
4. The domain of $1/x$ is every real number. _____

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $1/4 =$ _____ $1/(-2) =$ _____ $1/(1/5) =$ _____
2. Solve $1/x = 4$. $x =$ _____ Solve $1/x = -1$. $x =$ _____
3. Is there any x with $1/x = 0$? Explain.

4. As x grows very large and negative, what happens to $1/x$? _____

5. Fill in the pattern.

x	-0.1	-0.01	0.01	0.1
$1/x$				

Approaching 0 from the left the outputs go toward _____, from the right toward _____.

6. Why is $1/x$ the only parent function so far with a gap in its domain?

Exit Ticket

A function has no intercepts, a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, and two branches in Quadrants I and III.

Name it: _____ Domain in set notation: _____